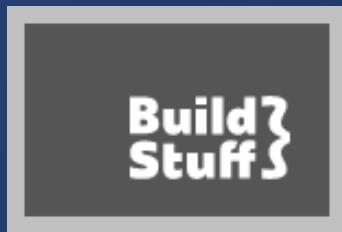


Intro to Blazing Web Accessibility

** With Microsoft Cognitive
Services*

2021 - Dennie Declercq



Welcome to a positive lifestyle!



Welcome on my website about a positive lifestyle. Too many people have stress that results in Burn-out, boredom and depression. I want to give you 3 easy advices for a positive and happy lifestyle. On this site you find advice on:

- Optimism
- Passion
- Purpose

Feel free to browse around for inspiration.

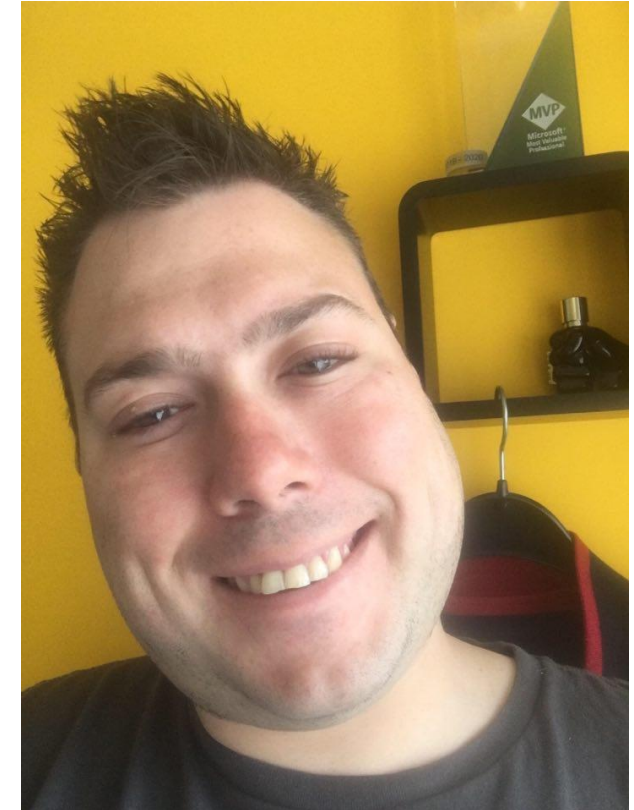
Introduction

- Introduction
- The Basics of Accessibility
- The Basics of Blazor
- The Basic of Cognitive Speech
- Introducing to The Future of Web Accessibility with Demo
- Step-By-Step Guide

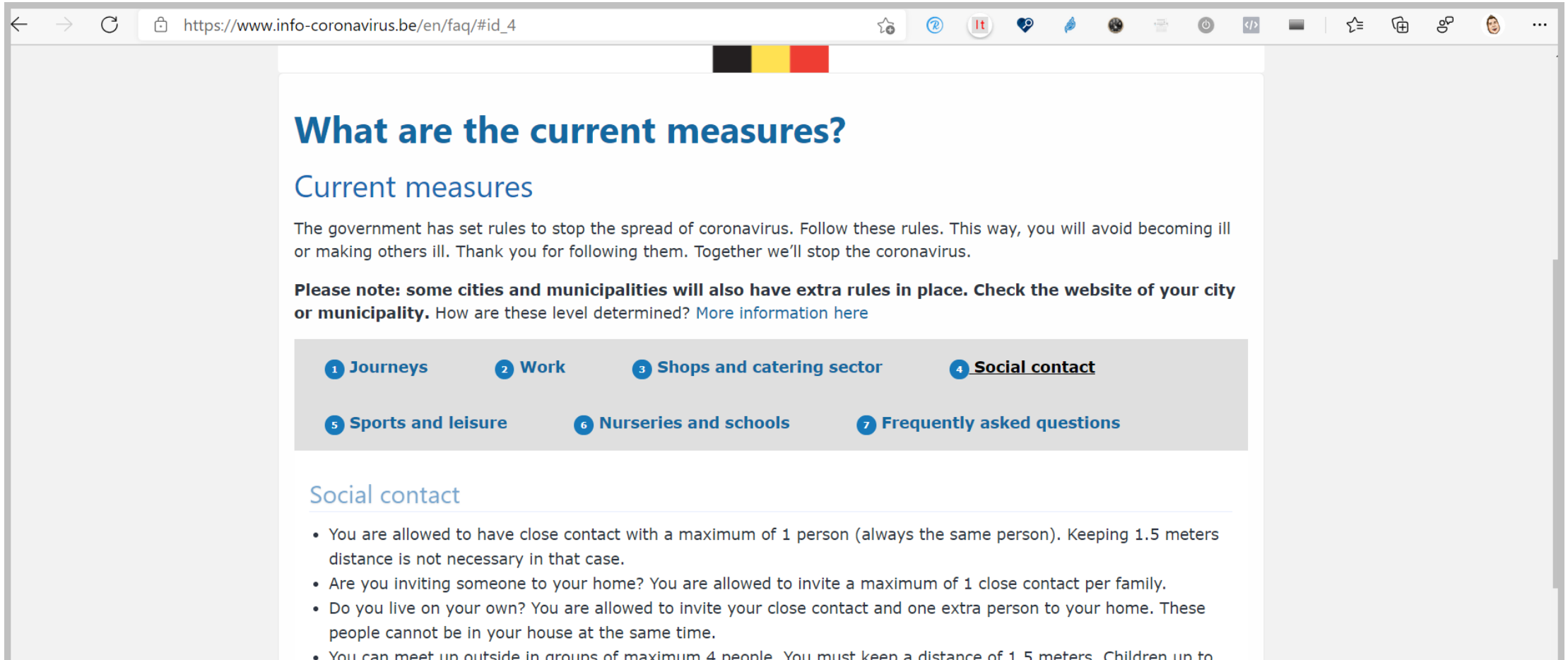
Content

Who am I?

- Dennie Declercq
- DDSoft
- President / Developer
- @DennieDeclercq
- #DreamingIsBelieving



2021: COVID19 – Inaccessibility - Exclusion



The screenshot shows a web browser window with the URL https://www.info-coronavirus.be/en/faq/#id_4. The page title is "What are the current measures?". Below the title is a section "Current measures" with a paragraph: "The government has set rules to stop the spread of coronavirus. Follow these rules. This way, you will avoid becoming ill or making others ill. Thank you for following them. Together we'll stop the coronavirus." A note follows: "Please note: some cities and municipalities will also have extra rules in place. Check the website of your city or municipality. How are these level determined? [More information here](#)". Below this is a horizontal menu with seven items: 1 Journeys, 2 Work, 3 Shops and catering sector, 4 Social contact (highlighted), 5 Sports and leisure, 6 Nurseries and schools, and 7 Frequently asked questions. Below the menu is a section "Social contact" with a list of bullet points: "You are allowed to have close contact with a maximum of 1 person (always the same person). Keeping 1.5 meters distance is not necessary in that case.", "Are you inviting someone to your home? You are allowed to invite a maximum of 1 close contact per family.", "Do you live on your own? You are allowed to invite your close contact and one extra person to your home. These people cannot be in your house at the same time.", and "You can meet up outside in groups of maximum 4 people. You must keep a distance of 1.5 meters. Children up to".

What are the current measures?

Current measures

The government has set rules to stop the spread of coronavirus. Follow these rules. This way, you will avoid becoming ill or making others ill. Thank you for following them. Together we'll stop the coronavirus.

Please note: some cities and municipalities will also have extra rules in place. Check the website of your city or municipality. How are these level determined? [More information here](#)

1 Journeys 2 Work 3 Shops and catering sector 4 **Social contact** 5 Sports and leisure 6 Nurseries and schools 7 Frequently asked questions

Social contact

- You are allowed to have close contact with a maximum of 1 person (always the same person). Keeping 1.5 meters distance is not necessary in that case.
- Are you inviting someone to your home? You are allowed to invite a maximum of 1 close contact per family.
- Do you live on your own? You are allowed to invite your close contact and one extra person to your home. These people cannot be in your house at the same time.
- You can meet up outside in groups of maximum 4 people. You must keep a distance of 1.5 meters. Children up to

Why Blazor and Cognitive Services?

Blazor

- Don't like JavaScript
- Never 'had JS under control'

Cognitive Services

- Possibilities that weren't possible
- True horse power

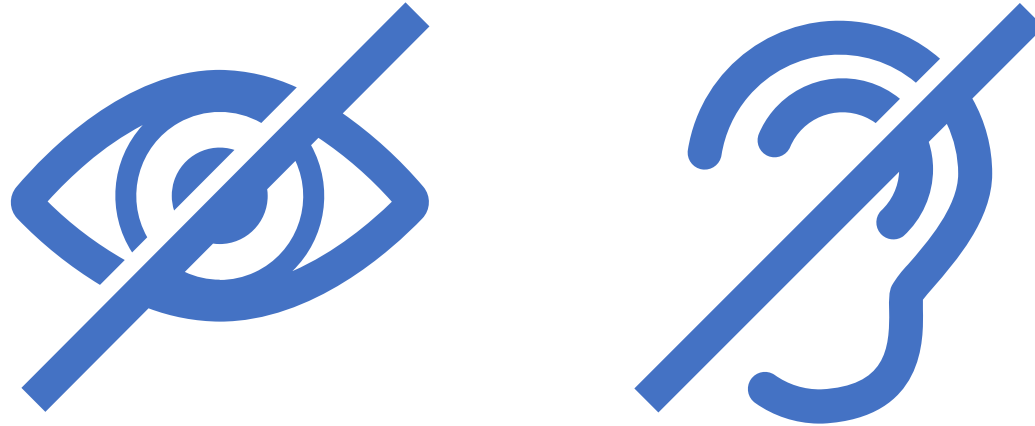
The basics of Accessibility



Difficulties

- Learning Disability
 - Reading
 - Understanding / Cognition
 - Structure
- Autism / Autism Spectrum Disorder
 - Structure
 - Confirmation
- Dyslexia
 - Reading
- Mental Health
 - Structure (sometimes)
 - Confirmation

Oh yes...



- And the well known Accessibility too
 - Blindness/ Vision loss
 - Auditory disability
 - Motoric restrictions

The Power of Text-To-Speech in Mobile apps



No, No, No, Narrator or Voice Over

Narrator / Voice Over

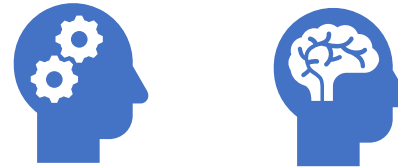
- Blind / Vision Loss



- How does it work?
- ➔ Reads sequentially whole screen

Text-To-Speech

- Learning Disability
- Dyslexia



- How does it work?
- ➔ Click a control and read aloud

The Basics of Blazor



What is Blazor?

- Microsoft's PWA/ Frontend language
- 2 Flavors:
 - Blazor Server Side (Hosted as .NET/ .NETCore Web App)
 - Blazor Web Assembly
- Affordable solution:
 - Azure Storage with Azure Static Web Apps

Getting started with Blazor

- VS 2021
- VS Code

Create a new Blazor app

.NET 5.0



Blazor Server App

A project template for creating a Blazor server app that runs server-side inside an ASP.NET Core app and handles user interactions over a SignalR connection. This template can be used for web apps with rich dynamic user interfaces (UIs).



Blazor WebAssembly App

A project template for creating a Blazor app that runs on WebAssembly. This template can be used for web apps with rich dynamic user interfaces (UIs).

Authentication

No Authentication

[Change](#)

Advanced Options

☒ Configure

☐ Enable

(Required)

Linux

Author: Microsoft

Source: GitHub

[Get additional project templates](#)

Exploring the Razor Syntax

```
page "/counter"

<h1>Counter</h1>

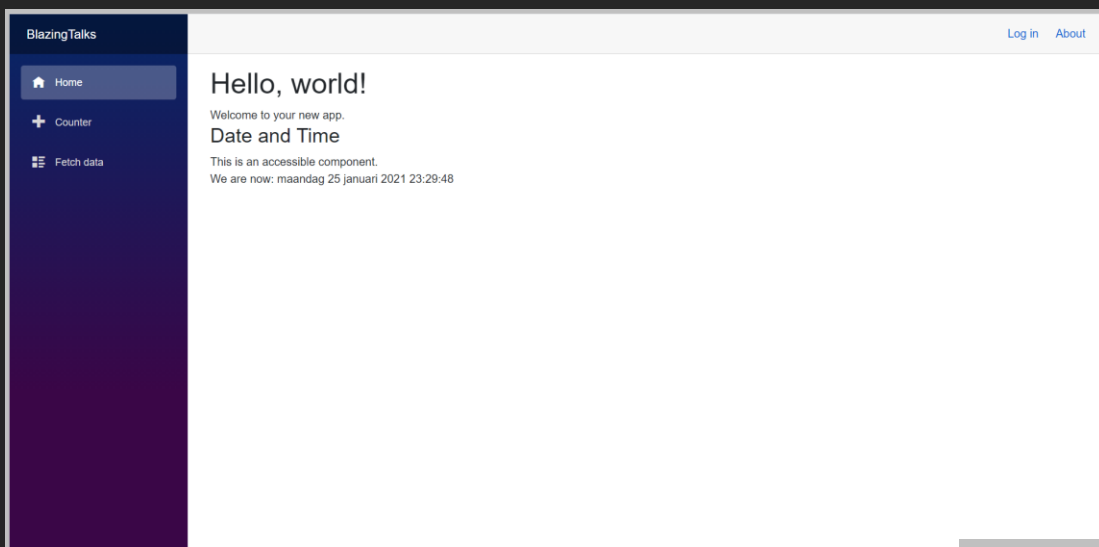
<p>Current count: @currentCount</p>

<button class="btn btn-primary" @onclick="IncrementCount">Click me</button>

@code {
    private int currentCount = 0;

    private void IncrementCount()
    {
        currentCount++;
    }
}
```

Razor components



```
@page "/"
```

```
<h1>Hello, world!</h1>
```

```
Welcome to your new app.
```

```
<AccessibleComponent />
```

```
<h3>Date and Time</h3>
```

```
<p>
```

```
    This is an accessible component. <br />
```

```
    We are now: @DateTimeString
```

```
</p>
```

```
@code {
```

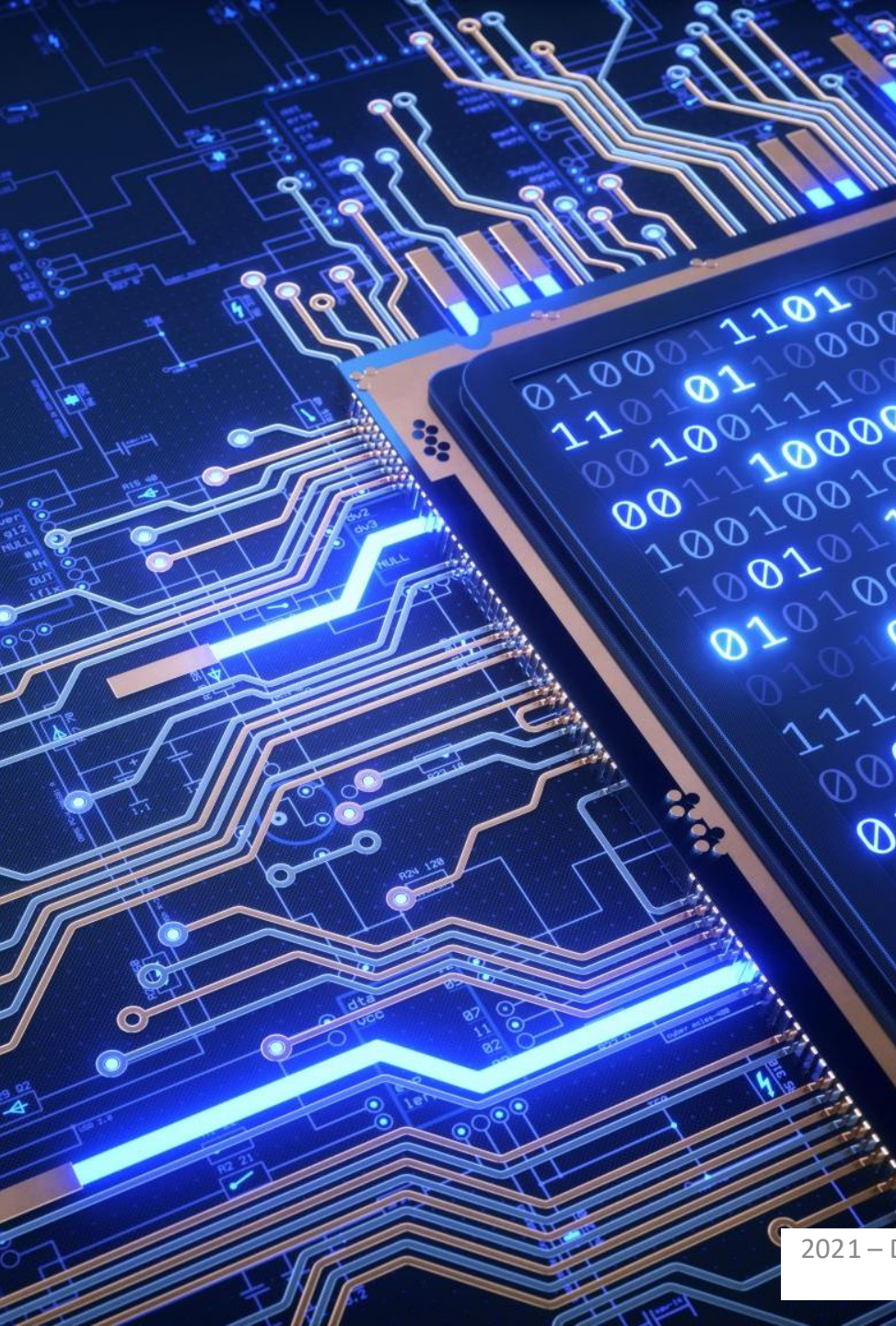
```
    string DateTimeString = DateTime.Today.ToLongDateString() + " " + DateTime.UtcNow.ToLongTimeString();
```

```
}
```


The Basics of Cognitive Services



 Microsoft
Cognitive Services



What is Microsoft Cognitive Services?

- Microsoft's AI platform
- Azure
- Pay as you go
- Developer focussed
- API Calls

SOME SERVICES:

AI + Machine Learning > Cognitive Services

 Bing Autosuggest Microsoft Help users complete queries faster by adding intelligent type-ahead capabilities	 Bing Custom Search Microsoft Create a custom search engine with ad-free results	 Bing Entity Search Microsoft Recognize and classify named entities, and find search results based on them	 Bing Search Microsoft Enable safe, ad-free search for your users with Bing, a leading search engine	 Bing Spell Check Microsoft Help users identify and fix word breaks, slang, names, homonyms, and brands	 Computer Vision Microsoft An AI service that analyzes content in images	 Content Moderator Microsoft An AI service that detects unwanted content
 Custom Vision Microsoft An AI service and end-to-end platform for applying computer vision to your specific scenario	 Face Microsoft An AI service that analyzes faces in images	 Speaker Recognition Microsoft An AI service that enables you to identify individual speakers or use speech as a means of verification	 Speech Microsoft An AI service that enables you to embed human-like speech capabilities into your apps	 QnA Maker Microsoft An AI service that enables you to create a bot from your data in minutes	 Text Analytics Microsoft An AI service that enables you to unlock insights from natural language text using sentiment	 Translator Text Microsoft An AI service that enables you to easily translate text between more than 60 languages
 Personalizer Microsoft An AI service that delivers a personalized user experience	 Language Understanding Microsoft Teach your apps to understand commands from your users	 Ink Recognizer (Preview) Microsoft An AI service that recognizes digital ink content, such as handwriting, shapes, and document layout	 Immersive Reader (preview) Microsoft An AI service that helps users read and comprehend text	 Form Recognizer (preview) Microsoft The AI service that extracts information from documents	 Anomaly Detector (preview) Microsoft An AI service that helps you foresee problems before they occur	 Cognitive Services Microsoft Connect powerful AI to your apps

Speech Service

- Text-to-speech
- Custom Voice



Introducing The Future of Web Accessibility

[Home](#)[Optimism](#)[Passion](#)[Purpose](#)

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A close-up photograph of a tiger's face, focusing on its eyes and the stripes on its forehead. The tiger has bright green eyes and a white stripe running down the center of its forehead. The background is dark and out of focus.

Warning

Demo

[Home](#)[Optimism](#)[Passion](#)[Purpose](#)

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Step-By-Step Guide to Implement

Snippets:

<https://go.ddsoft.be/gh-future-a11y>

Microsoft Cognitive Speech



 Microsoft
Cognitive Services

Creating a Speech Service

[Home](#) > [New](#) > [Marketplace](#) >

Speech



Microsoft



Speech

 [Save for later](#)

Microsoft

Create

Overview

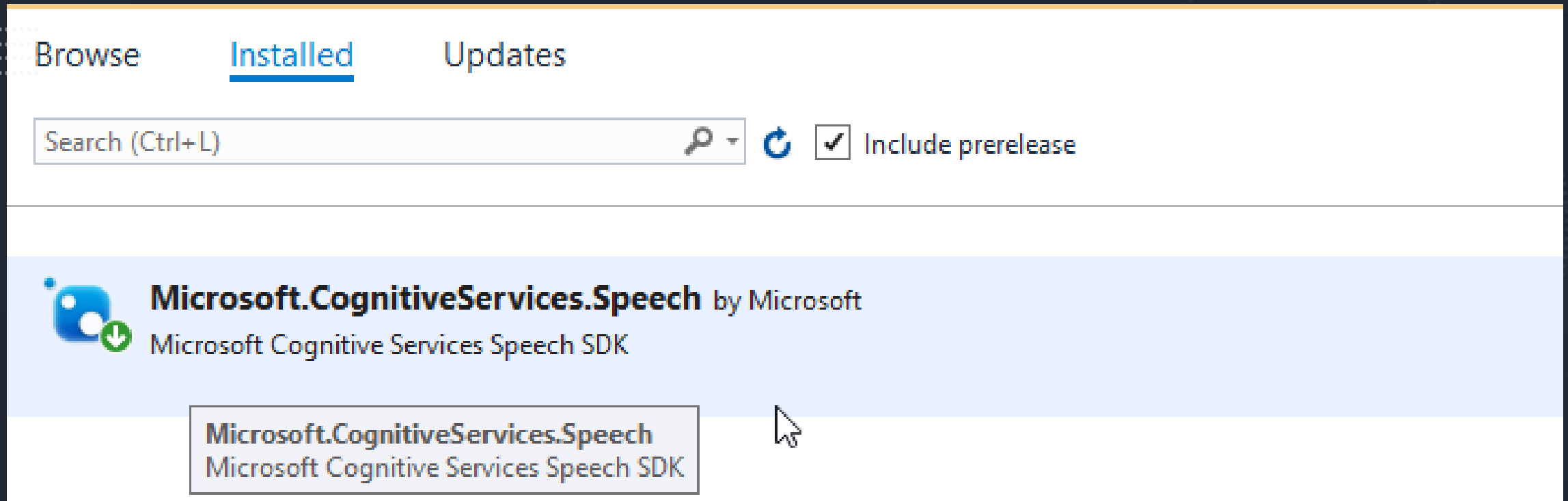
[Plans](#)

[Usage Information + Support](#)

Transcribe audible speech into readable, searchable text. Add real-time speech-to-text capabilities to your apps and services using the programming languages you are familiar with.

NuGet Package

Microsoft.CognitiveServices.Speech



Keys.cs

Public Static Class with connection keys to
Microsoft Cognitive Services

<https://github.com/DDSoftDennie/blazing-a11y-scripts/blob/code/cs/KEYS.cs>

CSSpeechService.cs

Code To call speech API

<https://github.com/DDSoftDennie/blazing-a11y-scripts/blob/code/cs/Services/CSSpeechService.cs>

JavaScript File

Select HTML element:

- Heading
- Paragraph
- ...

<https://github.com/DDSoftDennie/blazing-a11y-scripts/blob/code/js/interop.js>

PageCode



The diagram consists of two dark blue chevron-shaped boxes pointing to the right, connected by a white arrow. The first chevron contains the text '@inject IJSRuntime' and the second chevron contains the text '+ next Slides'.

@inject
IJSRuntime

+ next
Slides

Adding References to Elements

- Elements:
 - Headings
 - Paragraphs
 - ..
- Add: `@Ref="RefName" <EL @ref="RefName">Content</EL>`
- In `@Code` block:
 - Private ElementReferene RefName

<https://github.com/DDSoftDennie/blazing-a11y-scripts/blob/code/razor/DDA11yParagraph.razor>

ReadAloud method

Code to Read Aloud

<https://github.com/DDSoftDennie/blazing-a11y-scripts/blob/code/razor/DDA11yParagraph.razor>

EventHandlers

Code for the button

<https://github.com/DDSoftDennie/blazing-a11y-scripts/blob/code/razor/DDA11yParagraph.razor>

Walking over the Demo Solution (again)

Be aware of Basic HTML Accessibility

- Alt
- Usage of Headings
- Color contrast
- Aria
- Aria-hidden

Wrap up



Thanks || Q&A

-
- Dennie Declercq
 - dennie@ddsoft.be
 - @DennieDeclercq
 - #DreamingIsBelieving

